

Module 8: Convex Geometry — Answer Key

Prequiz

1. The filled rectangle is convex. A line segment between any two points in it stays in the rectangle. A crescent has pairs of points whose connecting segment leaves the set.
2. Boundary $x + 2y = 8$ has intercepts $(8, 0)$ and $(0, 4)$. For $(2, 3)$, $2 + 2(3) = 8$, so it is feasible. For $(4, 3)$, $4 + 6 = 10 > 8$, so it is not feasible.
3. Solve

$$x + y = 8, \quad 2x + y = 10.$$

Subtract to get $x = 2$, hence $y = 6$. The vertices are

$$(0, 0), (5, 0), (2, 6), (0, 8).$$

4.

$$P(0, 0) = 0, \quad P(5, 0) = 20, \quad P(2, 6) = 38, \quad P(0, 8) = 40.$$

$$\text{Maximum} = 40 \text{ at } (0, 8).$$

5.

$$f(0, 0) = 0, \quad f(6, 0) = 18, \quad f(4, 3) = 18, \quad f(0, 5) = 10.$$

The adjacent vertices $(6, 0)$ and $(4, 3)$ tie, so every point on the edge joining them is optimal.

Matching quiz

1. The filled triangle is convex; the ring is not because some connecting segments pass through the hole.
2. Boundary $3x + y = 12$ has intercepts $(4, 0)$ and $(0, 12)$. At $(3, 4)$, $3(3) + 4 = 13 > 12$, so the point is not feasible.
3. Solve

$$x + y = 7, \quad 2x + y = 10$$

to get $(3, 4)$. The vertices are

$$(0, 0), (5, 0), (3, 4), (0, 7).$$

4.

$$R(0, 0) = 0, \quad R(5, 0) = 25, \quad R(3, 4) = 31, \quad R(0, 7) = 28.$$

$$\text{Maximum} = 31 \text{ at } (3, 4).$$

5. Every point on the edge joining the two adjacent optimal vertices is also optimal.